

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

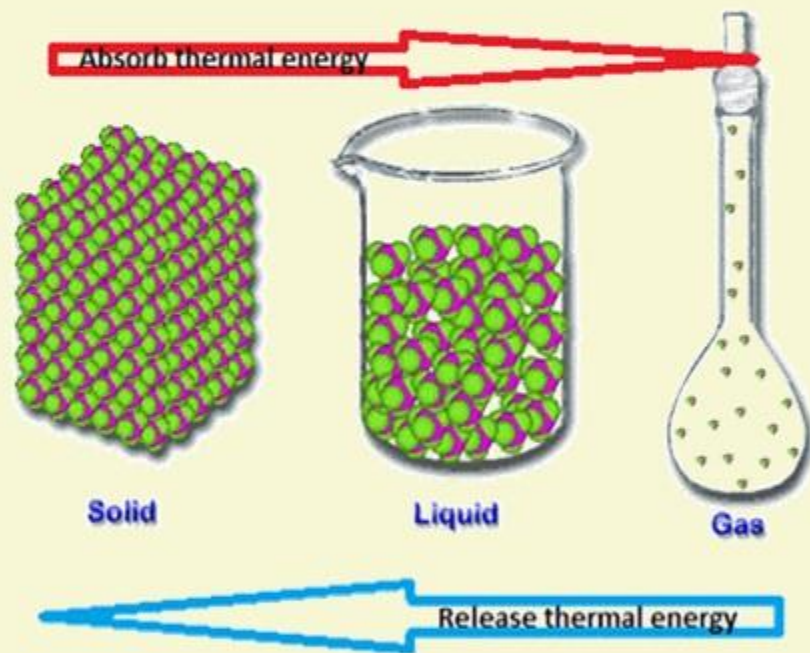
MODULE – 26

- HUMIDITY
- DEW POINT
- FOG
- MIST & HAZE

CLIMATOLOGY MODULES

Atmosphere	▪ Composition and structure; weather and climate;
Insolation	▪ Factors controlling insolation; heat budget of the earth — heating and cooling of atmosphere (conduction, convection, terrestrial radiation, advection);
Temperature	▪ Factors controlling temperature; distribution of temperature — horizontal and vertical; inversion of temperature;
Pressure	▪ Pressure belts; winds-planetary, seasonal and local; air masses and fronts; tropical and extra-tropical cyclones;
Precipitation	▪ Evaporation; condensation — dew, frost, fog, mist and cloud; rainfall — types and world distribution;
World Climate and Climatic Changes	▪ Different types of Climate

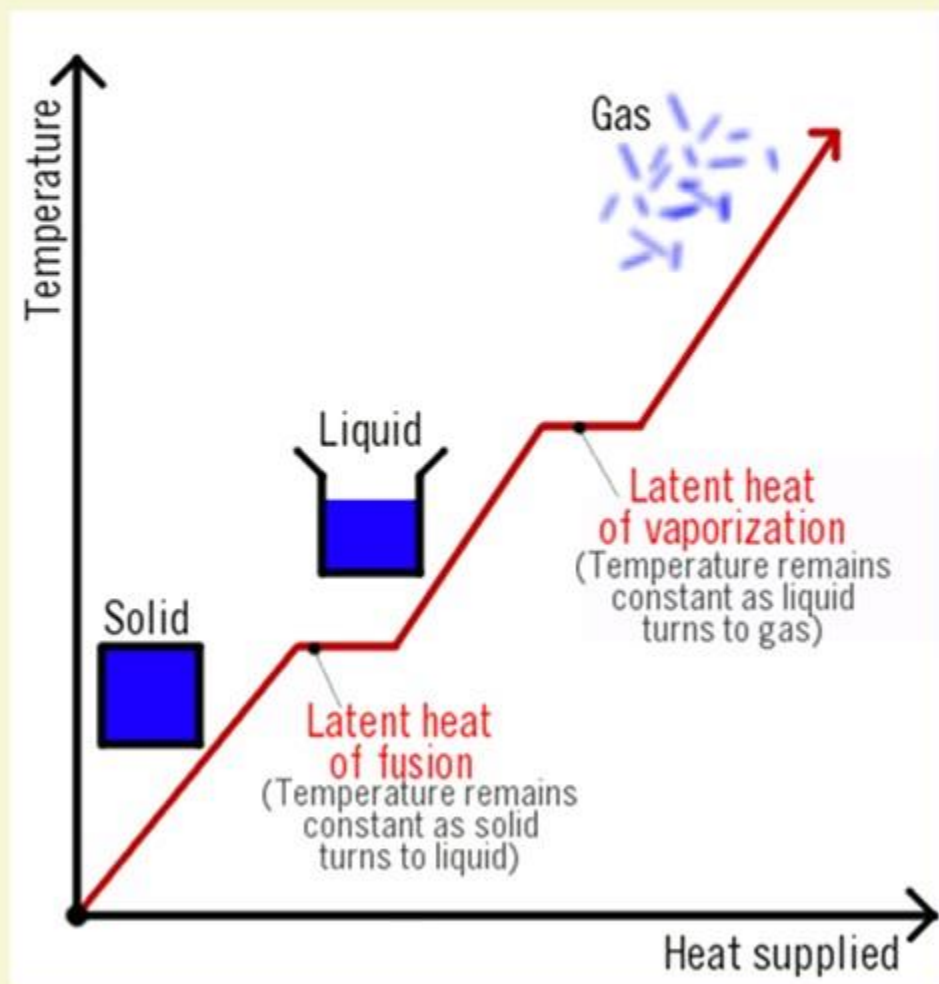
STATES OF WATER AND HEAT EXCHANGE



1 Water occurs in three states: solid as ice (frozen), liquid as water and gaseous as water vapour.

2 When water passes into vapour, the sensible heat passes into hidden form held by water vapour. The **energy is absorbed in this case** and this is known as the **latent heat of vaporisation**.

STATES OF WATER AND HEAT EXCHANGE



3 Likewise, the heat needed in the process of melting to change a substance from a solid at its melting point into liquid at the same temperature is known as **latent heat of fusion**.

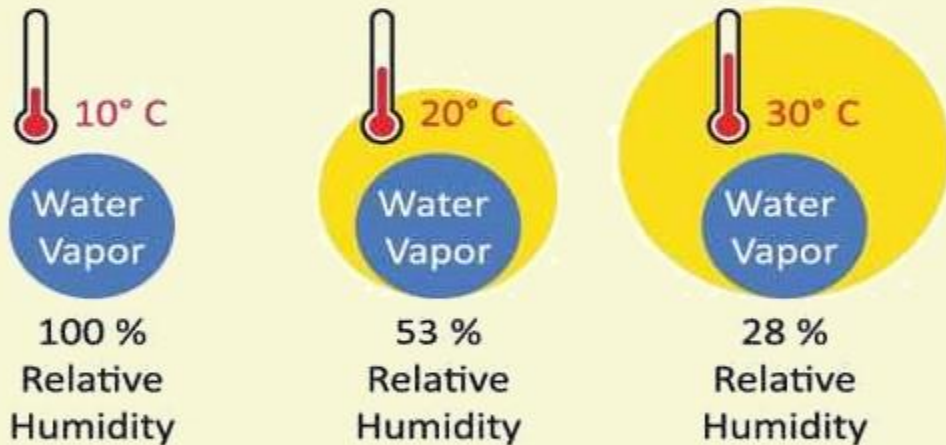
4 In case of **sublimation**, the amount of heat energy involved is far greater as it includes both the latent heats of vaporisation and fusion.

HUMIDITY



- 1 The term humidity refers to the amount of water vapour present in the air.
- 2 For any specific temperature there is a limit to the quantity of water vapour that can be held by the air.
- 3 This limit is called the saturation point.

RELATIVE HUMIDITY



- 1 The proportion of **water vapour** present relative to the maximum quantity at a given **temperature** is called **relative humidity** and is expressed as a percentage.
- 2 Saturation point thus means a relative humidity of 100 percent.
- 3 The relative humidity can be changed either by increasing or reducing the amount of **water vapour** present in the air or by **changing the temperature** of air.
- 4 Lowering the temperature of air without changing the amount of water vapour present in it will increase the relative humidity while an **increase in its temperature** will result in a reduced relative humidity.

OTHER EXPRESSIONS OF HUMIDITY

1 There are also some other expressions used to denote humidity.

2 ABSOLUTE HUMIDITY:

The term Absolute Humidity denotes the **actual quantity of water vapour present in air** and it is defined as the weight of water vapour (grams) in a given volume of air (cubic metre)

$$\text{Absolute Humidity} = \frac{\text{Weight of vapour}}{\text{Volume of air}}$$

- *Humidity is measured by an instrument called the hygrometer.*
- *Hygograph is an instrument which plots a graph of relative humidity round the clock.*

3 SPECIFIC HUMIDITY:

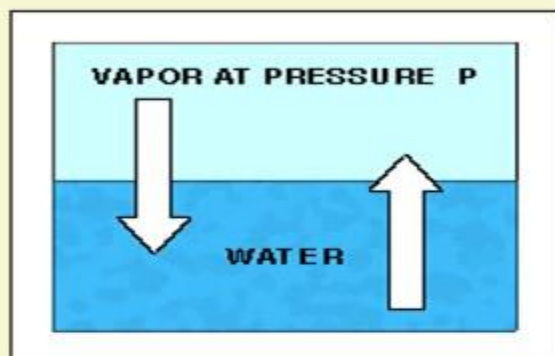
The term Specific Humidity is applied to express the **ratio of weight of water vapour to the weight of moist air** (including water vapour). It is stated as grams of water vapour per kilogram of moist air.

$$\text{Specific Humidity} = \frac{\text{Weight of Vapour}}{\text{Weight of Moist air}}$$

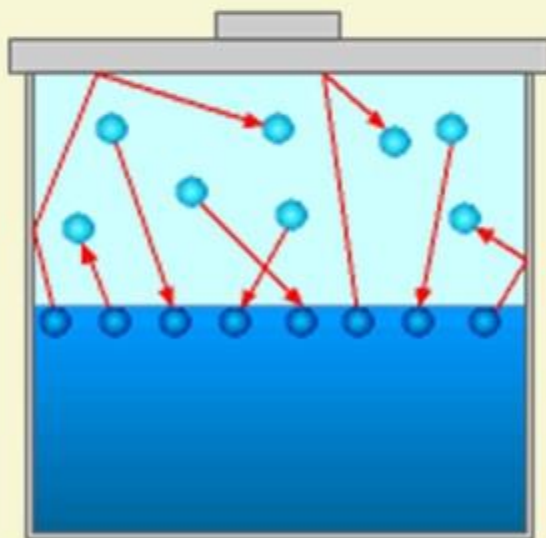
4 MIXING RATIO:

The term Mixing Ratio is used by meteorologists. It is the **ratio of weight of water vapor to the weight of dry air** and is stated as grams per kilogram.

VAPOUR PRESSURE



CHAMBER AT TEMPERATURE T



- 1 The column of air at any point exerts a pressure. The amount of pressure exerted by a given column of air differs according to the varying amounts of water vapour present in it.
- 2 That part of barometric pressure due to water vapour alone is known as vapour pressure.
- 3 The maximum vapour pressure at any temperature occurs when the air is saturated.

DEW POINT AND CONDENSATION

- 1 The term condensation is applied to the process of the change of state of water from vapour to liquid.



- 2 This is the basis of all kinds of precipitation, the fall of water from the atmosphere to the ground in any form.
- 3 The term **Dew Point** is applied to that **critical temperature** at which air gets fully saturated with the present amount of moisture.
- 4 In other words this is the temperature at which the relative humidity of the concerned parcel of air will become **100 percent** without any addition of water vapour to it.
- 5 Condensation usually follows if the temperature of the air falls below the dew point.

DEW AND FROST



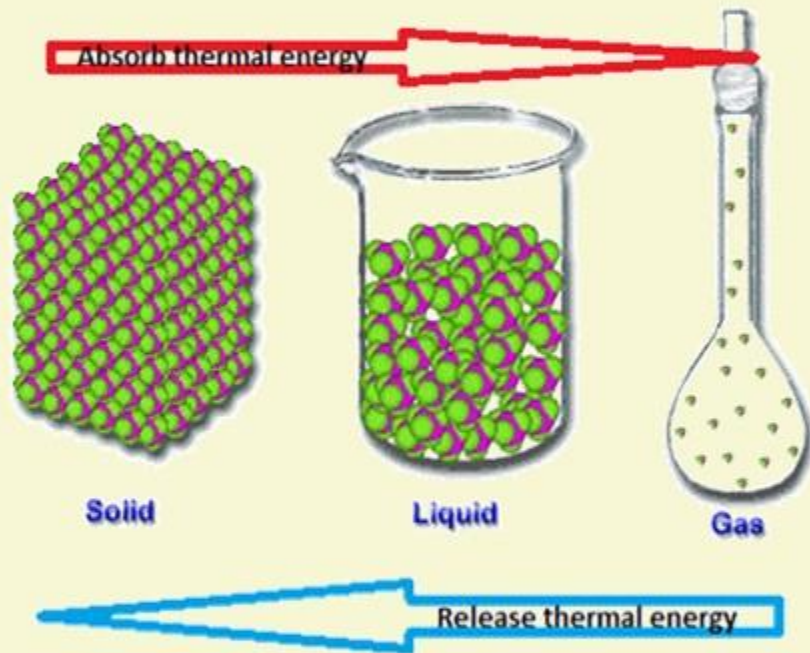
- 1 At night, the Earth surface and various objects such as leaves etc. may frequently become cooler than the air above it.
- 2 This causes water vapour contained in the air to get condensed and deposit on the cooled surface. This is known as dew.
- 3 The ideal conditions for its formation are clear sky, calm air, high relative humidity and cold and long nights.
- 4 On cloudy nights cooling of the surface is not complete due to the greenhouse effect of the clouds and dew is not likely to be formed.

DEW AND FROST



- 5 In winter season the cooling of the surface may cause the ground temperature to fall below the freezing point.
- 6 In such a situation the moisture contained in air condenses and freezes on various objects on the ground making a thin layer of ice. This is called **Frost**.
- 7 Frost is harmful to crops and irrigating the fields is attempted as a precaution against it. Small and tender plants can also be protected from frost by keeping them covered at night.

IRRIGATION PROTECTS THE CROPS FROM FROST!



DO U KNOW HOW?

Water to turn into ice it must give up (or lose) heat. It is this heat that is lost during the freezing process that keeps the plants from being damaged!

FOG

Fog can be defined as a **very low stratus cloud** that has **formed close to the ground**. On the basis of its formation fog can be of various types.

1. RADIATION FOG

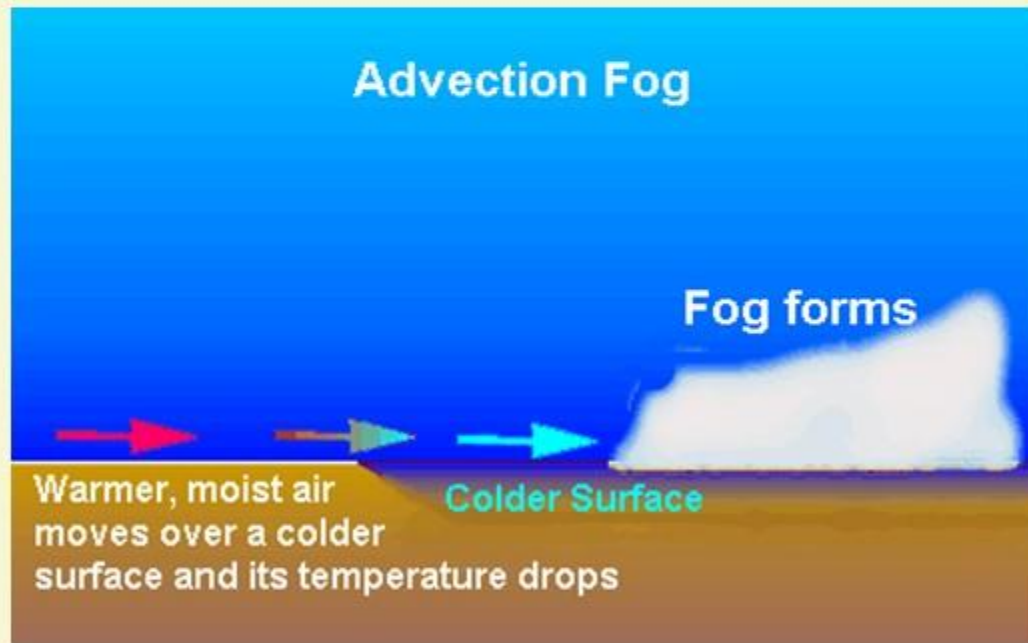


1 Radiation fog is usually associated with **temperature inversion** and is formed at night when **the temperature of air near the ground falls below dew point**.

2 The moisture in the air gets condensed and small particles of water remain suspended in the air reducing visibility.

FOG

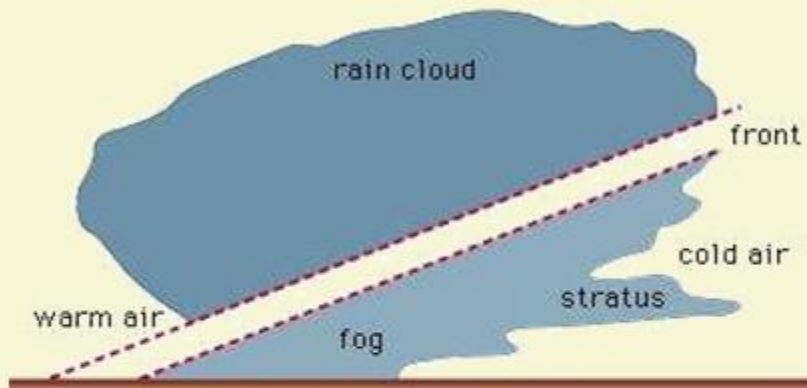
2. ADVECTION FOG



- 1 Advection fog, another type of fog, develops when **warm moist air blows over a cold surface such as a snow covered area.**
- 2 As the air gets cooled when it comes in contact with the cold surface, water vapour contained in it gets condensed and forms the fog.
- 3 Many of the **coastal regions** also experience advection fog.
- 4 This type of fog occurs when the **relatively warmer and humid air of maritime origin** moves towards the coast and gets cooled as it blows over the colder continental areas.

FOG

3. FRONTAL FOG



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- 1 Frontal fog is a result of the mixing of warm and cold air masses. In such zones of convergence the warm air has a tendency to rise over the denser cold air .
- 2 On uplift and due to its interaction with the cold air the warm air gets cooled and the moisture in it gets condensed forming fog.
- 3 Fog of New Foundland region in the Northern Atlantic Ocean is of this type

SMOG



- 1** Smog is also called smoke fog. Smog is a combination of water particles and fine particles of carbon and other substances contained in smoke.
- 2** Smog occurs in areas containing large amount of smoke, for example industrial areas.
- 3** It develops more quickly than the fog and lasts longer.

MIST



- 1 Mist is also a type of fog and the visibility in this case is between 1 to 2 km.
- 2 In case of mist, water droplets occur in the lower layers of the atmosphere.
- 3 The only difference between the mist and the fog is that mist contains more moisture than fog.
- 4 Fogs, on the other hand, are mini clouds in which condensation takes place around nuclei provided by the dust, smoke and the salt particles.

HAZE



- 1** Haze is the reflection of sunlight off air pollution. The difference between haze and smog is that smog is what happens when pollution causes low-lying ozone.
- 2** Haze may be produced by presence of dust also which reduce visibility.
- 3** Haze due to dust particles in air is common in tropical regions during dry season.

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